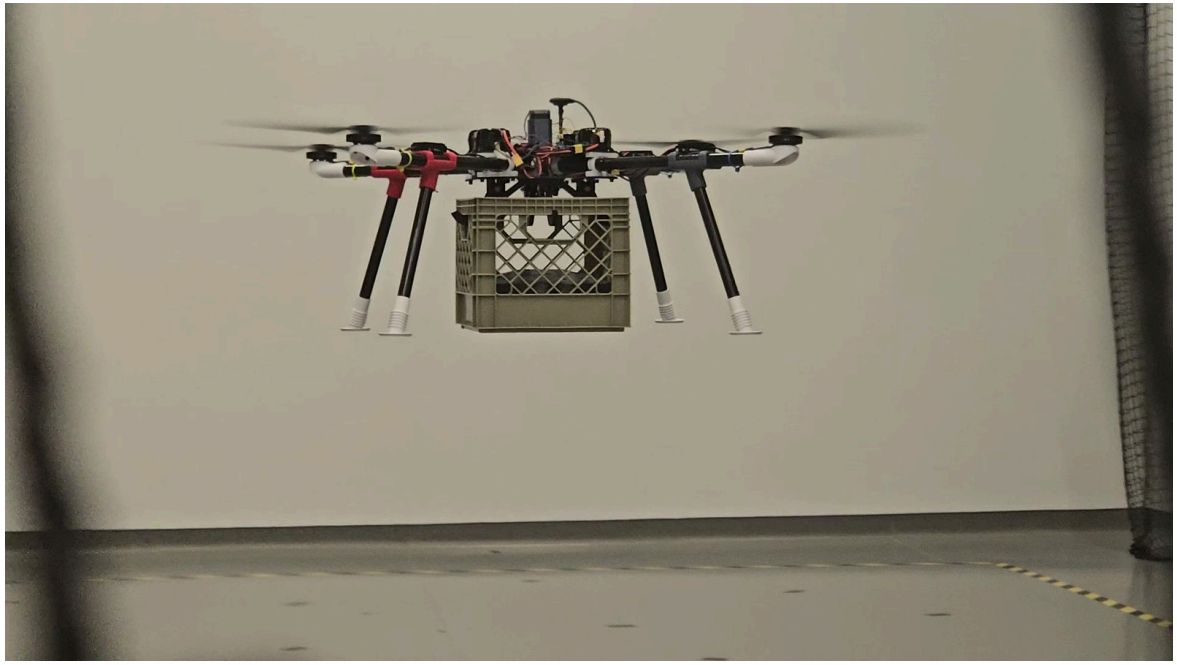


Heavy Lift Drone

Electrical subsystem lead on a multirotor rated to lift 30 lb: power distribution, harnessing, motor control, and a lighter arm coupling.

I led the electrical subsystem of a heavy lift multirotor: the power distribution board, the hand built harness, and motor control. I also redesigned the coupling that joins the carbon fiber arm tubes, cutting it from 10 parts to 6 and its weight by 14%. In April 2026 the drone flew with a payload in the test cage.

Context ASME Design Team, Penn State	Dates Fall 2025 to April 2026	My role Electrical subsystem lead
Scope Power distribution, cable fabrication, motor control, arm coupling	Tools SolidWorks, SolidWorks Simulation, force gauge	Outcome Flew with a payload in cage testing

30 lb
rated lift

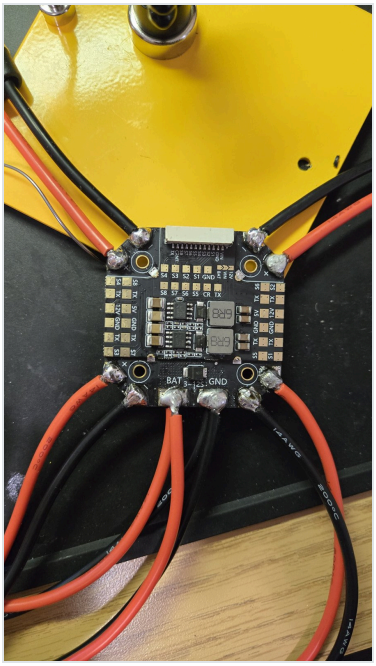
10 to 6
parts in the redesigned arm coupling

14%
lighter coupling

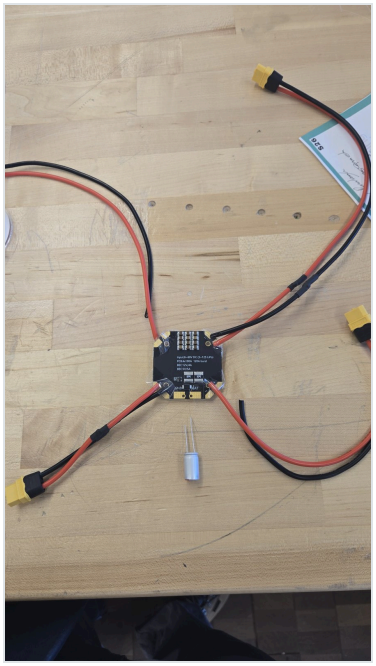
Project	Sheet title	Engineer	Date	Sheet
Heavy Lift Drone	Overview	Xiang Wang	2025 to 2026	1 of 5

Power distribution and harness

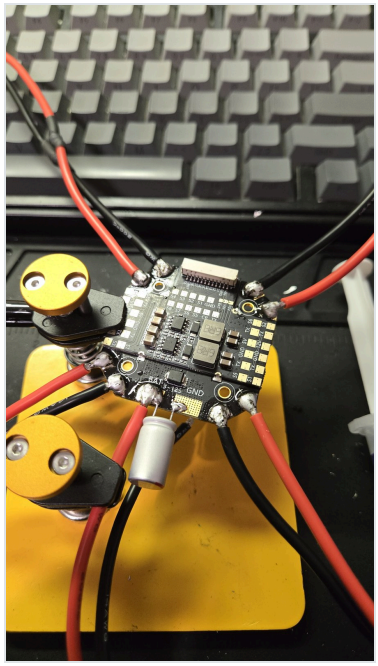
Every ESC draws from a central power distribution board. I laid the harness out on the bench at full length, fabricated the cables and connectors, and checked the system on the bench before it went on the airframe.



2.1 Power distribution board with ESC leads



2.2 Harness laid out at length

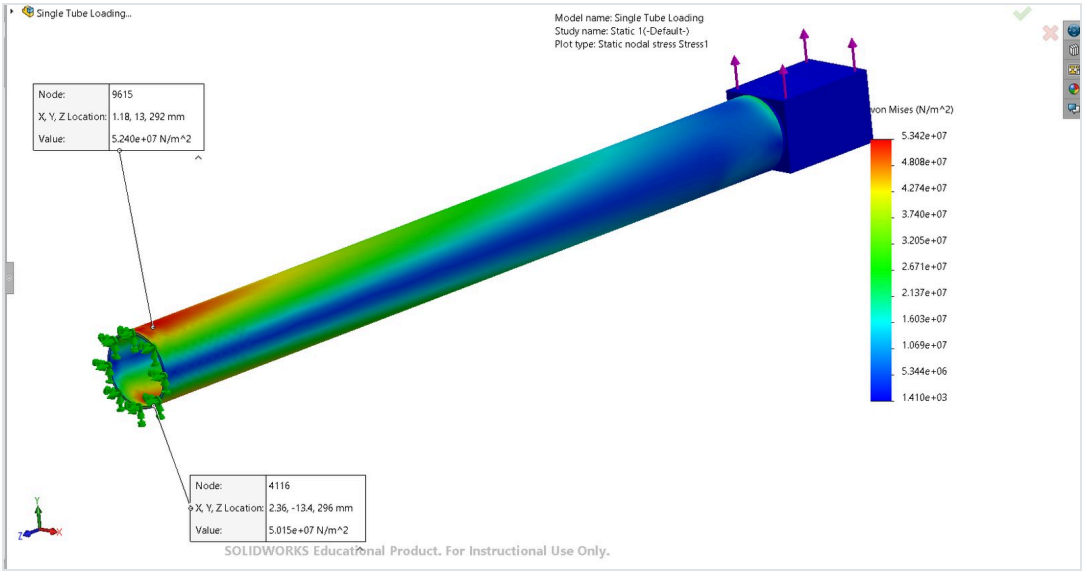


2.3 Board installed on the frame

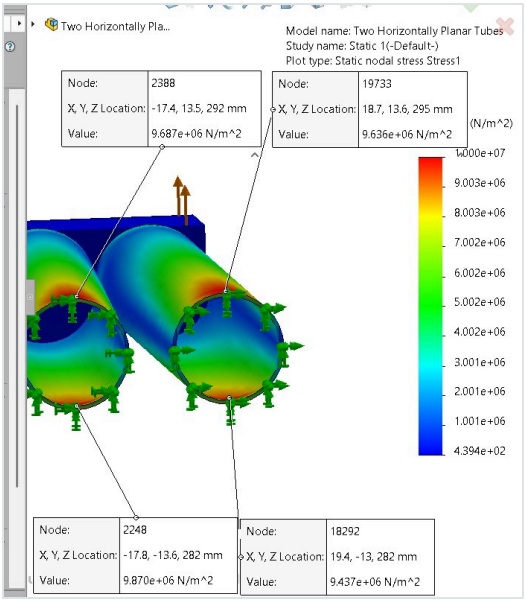
Project	Sheet title	Engineer	Date	Sheet
Heavy Lift Drone	Power distribution and harness	Xiang Wang	2025 to 2026	2 of 5

Arm structure

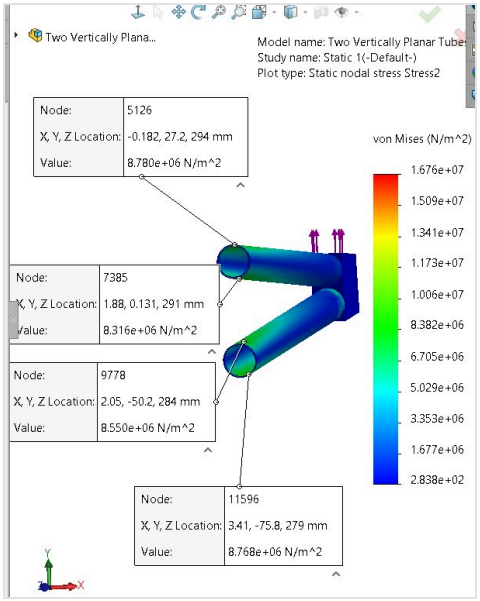
The arms are carbon fiber tubes. Before settling the layout I compared a single tube arm against paired tubes stacked horizontally and vertically in SolidWorks Simulation, loading the motor end and probing von Mises stress at the root.



3.1 Single tube arm under tip load



3.2 Paired tubes, horizontal



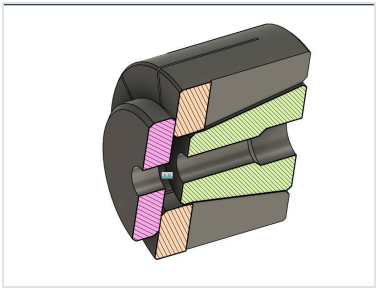
3.3 Paired tubes, vertical

Project	Sheet title	Engineer	Date	Sheet
Heavy Lift Drone	**Arm structure**	**Xiang Wang**	**2025 to 2026**	**3 of 5**

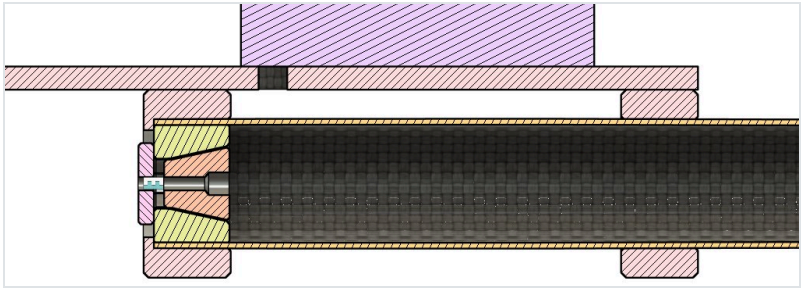
Coupling redesign

The original coupling for the carbon fiber arm tubes used 10 components. I replaced it with a slotted collet clamp, bringing it down to 6 components and cutting its weight by 14%.

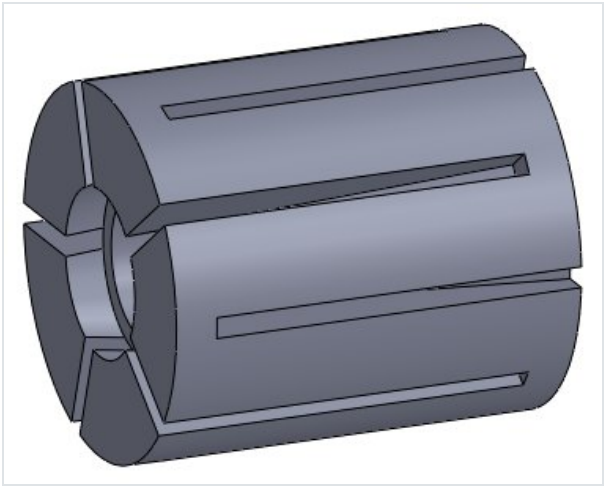
The collet was pull tested on an arm tube with a force gauge before going on the airframe.



4.1 Collet coupling, section view



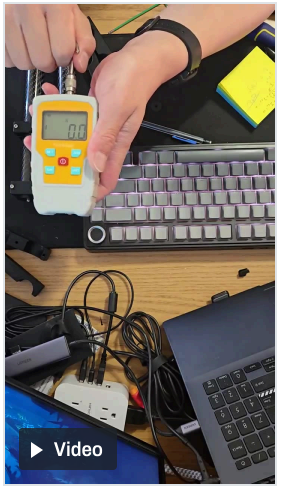
4.2 Collet clamping the carbon tube



4.3 Slotted collet



4.4 Printed collet



4.5 Pull test on an arm tube with a force gauge

Project	Sheet title	Engineer	Date	Sheet
Heavy Lift Drone	Coupling redesign	Xiang Wang	2025 to 2026	4 of 5

Flight testing

Parts were load checked on the bench with a force gauge before flight. In April 2026 the drone lifted a crate payload and hovered inside the netted test cage.

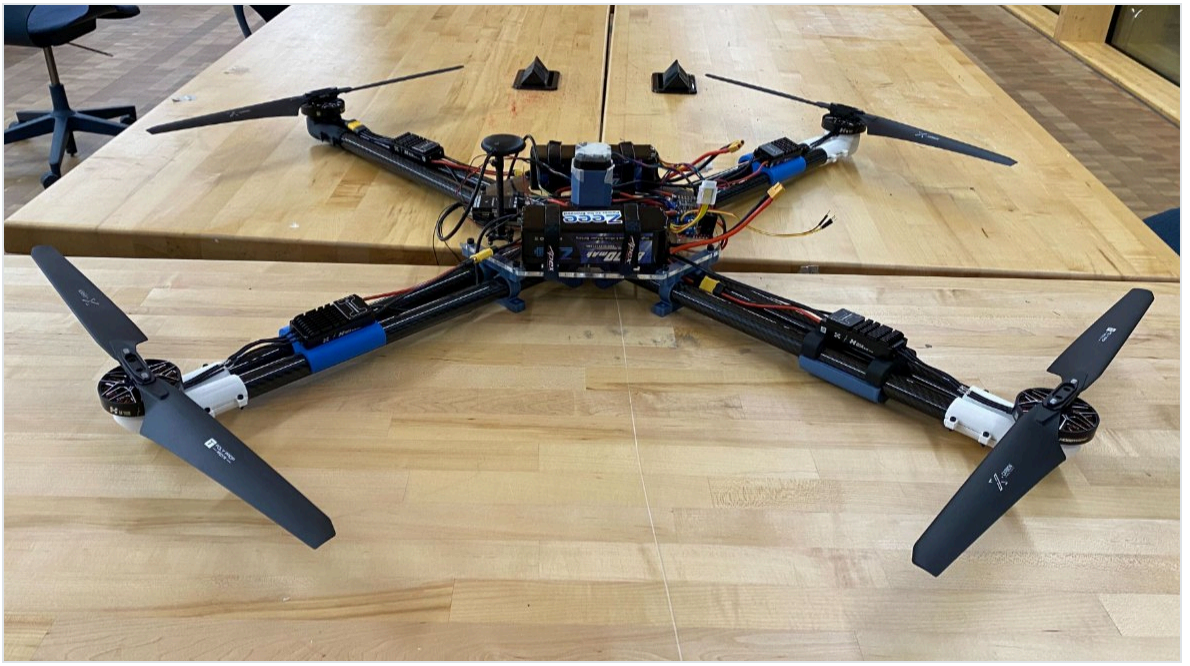
Files

Case study (PDF)

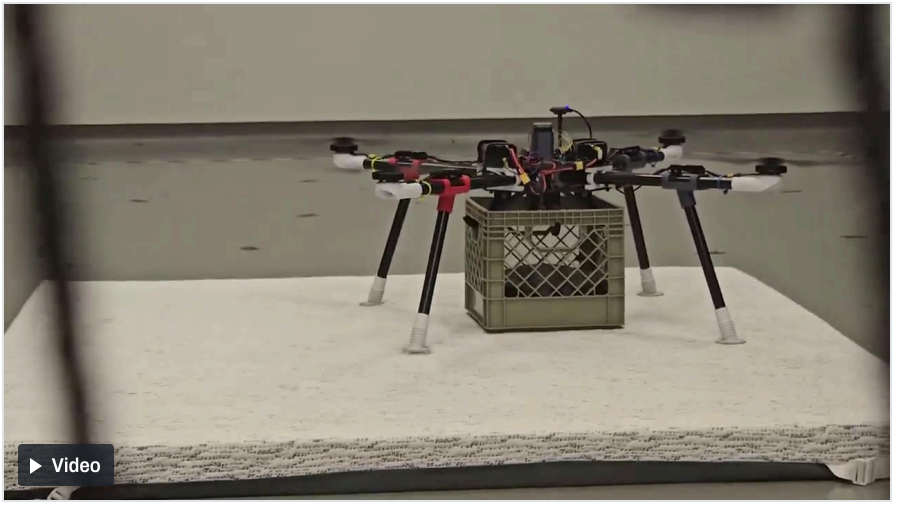
Drone assembly (STEP)

github.com/xiang-wang-US/Heavy_Lift_Drone

Built with the ASME Design Team at Penn State.



5.1 Airframe before payload testing



5.2 Payload lift in the test cage, April 2026



5.3 Bench load check with a force gauge

Project	Sheet title	Engineer	Date	Sheet
Heavy Lift Drone	Flight testing	Xiang Wang	2025 to 2026	5 of 5